1. Top 10 Protocol Selection

The experiment uses 10 major Ethereum ecosystem protocols:

1. Aave
2. Balancer
3. Compound
4. Curve
5. ENS
6. EigenLayer
7. Lido
8. Morpho
9. Pendle
10. Uniswap

For each protocol, official Ethereum Mainnet production/core contract addresses were collected from official documentation or official project repositories.

Seed Dataset:

```
data/input/top_10_addresses.csv
```

Layer 1 Discovery

Data Source

Layer 1 address discovery uses the Alchemy Ethereum Mainnet Transfers API. by:

```
alchemy_getAssetTransfers
```

Pagination

Alchemy's Transfers API can return multiple pages of results.

Script supports pagination using. `pageKey` of Alchemy's

intentionally something like:

```
Request page ---> Receive transfers ---> pageKey ?
                                            |
                                            |--- No -> Finished
                                            |
                                            |--- Yes
```

```
                                                  |
                                          Request next page
```

But for the current demonstration, the traversal is intentionally limited to:

```
2 pages per direction per seed contract
```

The purpose of this limit is to demonstrate the methodology, measure graph growth, and avoid performing a large historical extraction before deciding whether deeper traversal is worthwhile.

Output

```
data/output/top_10_related_addresses.csv
```

```
data/output/layer1_growth_summary.csv
```

Growth by Protocol:

Protocol	Seed Contracts	Relationship Edges	Unique Related Addresses
Aave	2	63	45
Balancer	1	400	32
Compound	2	639	208
Curve	1	10	3
ENS	2	298	63
EigenLayer	4	1,055	537
Lido	2	402	303
Morpho	1	400	44
Pendle	1	374	139
Uniswap	4	859	215
Total	27	4,500	1,572

Project Structure

```
ethereum-top10-analysis/
|
```

```
|--- .venv/
|--- data/
|    |--- input/
|    |    |--- top_10_addresses.csv
|    |
|    |--- output/
|         |--- layer1_growth_summary.csv
|         |--- top_10_related_addresses.csv
|
|
|--- src/
|    |--- layer1_discovery.py
|
|--- .env
|--- .gitignore
|--- README.md
|--- requirements.txt
```